

Paramount+ · Subscriber Regression Model

Linear Regression: Predicting Total Signups · Feb–Apr 2026

Channel Efficiency Outperforms Spend

Efficient budget allocation and marketing strategy both have a greater impact on sign-ups. The data shows that average return on ad spend (ROAS) is the strongest predictor of sign-up volume. The goal of this linear regression model was to determine whether sign-ups are influenced by ad spend, plan price, subscriber age, and bundling. By identifying which channels produce the highest ROAS, Paramount+ can focus investment where it drives the most subscriber growth.

Model Performance

R-squared

0.958

95.8% variance explained

RMSE

2.98

±3 signups avg error

Target (y)

Total Signups

Campaign-level prediction

Train / Test

80% / 20%

Random state = 42

Feature Importance (by Coefficient)

Feature

1. ROAS 287.34 ★★★★★

Higher ROAS leads to significantly more signups.

2. Bundled +0.020 ★★★★★

Bundled subscribers are slightly associated with more signups.

3. Age +0.002 ★★★★★

Minimal positive effect from subscriber age.

4. Total Spend +0.0009 ★★★★★

More spend leads to marginally more signups, therefore efficiency matters more than volume.

5. Plan -0.0005 ★★★★★

Higher price plan does have a slight negative effect on signups.

Correlation with total_signups

Feature

r value

Strength

Direction

Interpretation

total_spend

0.837

Strong

Positive ↑

Highest correlator: spend volume drives signup volume

avg_roas

0.223

Moderate

Positive ↑

Better efficiency correlates with more signups

plan_price

0.010

Very Weak

Positive ↑

Plan pricing has almost no effect on signups

age

0.006

Negligible

Positive ↑

Subscriber age does not predict signups

is_bundled

-0.054

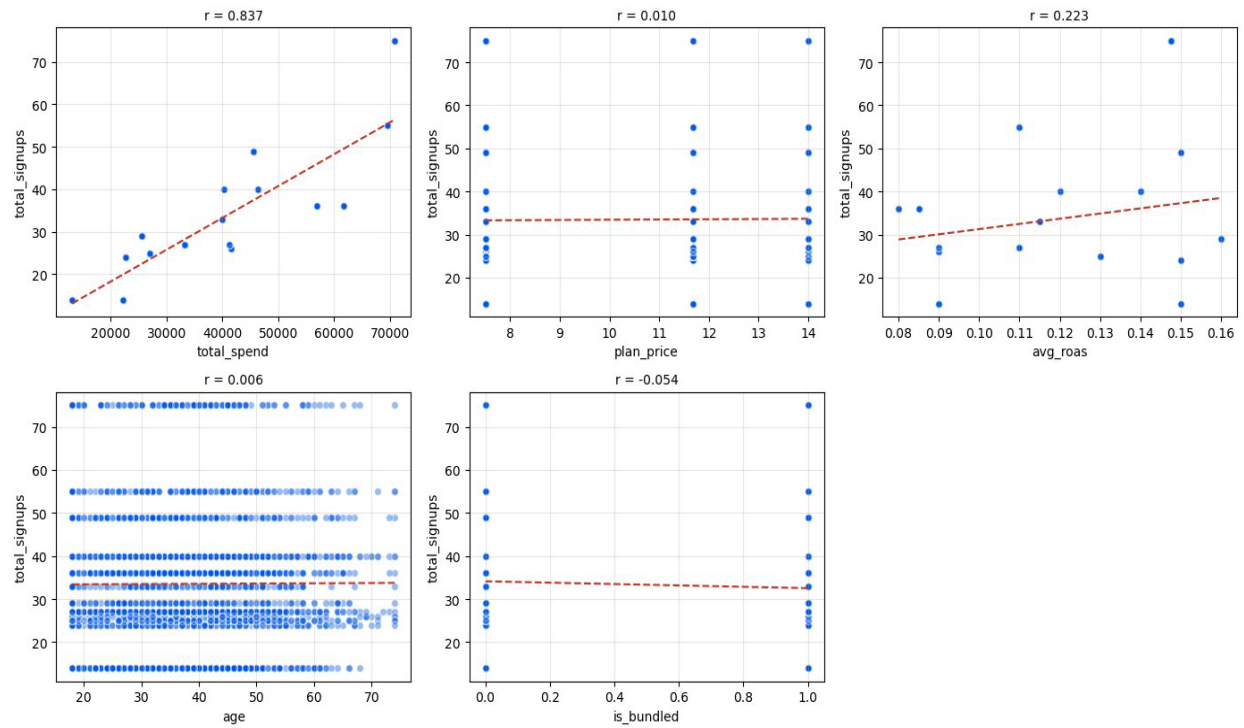
Very Weak

Negative ↓

Bundled subs slightly associated with fewer signups

Scatter Plots — Feature vs Total Signups

Feature Correlations with Total Signups



Process & Data Limitations

Process

Generated synthetic data in Claude AI before cleaning and preprocessing data in Python using Pandas train/test splitting. Applied an 80/20 train/test split and fit a multivariate linear regression model using scikit-learn.

Data Limitations

A production model would incorporate adstock decay, seasonality, and 18–24 months of historical spend data to improve predictive accuracy. Statistical significance testing between campaign groups would further validate the regression findings.